A Constructive Introduction to Quantum Field Theory

Lecture 6: Φ^4_3 on the torus II. A priori estimates for the forward–backward system

BY M. GUBINELLI

Mathematical Institute, University of Oxford

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Abstract

We prove the central estimate of the construction: bounds, uniform in the lattice, on the error Z of the approximate forecast and on the time the forecast spends at large fields. The two halves of the FBSDE of Lecture 5 give two inequalities, neither of which closes alone. The forward equation, rewritten as a cubic elliptic equation with memory, is coercive: a convex threshold test bounds the weighted large-field action by Gaussian tails and by the backward energy. The backward equation is linear in Z: a conditional representation formula bounds the backward energy by an integrable bounded-field source, a boundary term, and a *small* multiple of the large-field action. The small coefficient is $C\gamma^2/q_0$, independent of the lattice and of the moment order, so one choice of q_0 closes the loop. The clipping error of Lecture 5 is controlled exactly in this way. We give complete proofs in the scalar (one-site) model, where all the mechanisms are visible with explicit constants, and indicate the changes needed on the lattice.

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Introduction

At the end of Lecture 5 we had, at each lattice cutoff N, the exact system (Theorem 10.6)

$$\begin{aligned} \mathrm{d}Y_q &= [b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + H_q(Y_q)\,\mathrm{d}W_q, \\ \mathrm{d}Z_q &= -[b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + \Xi_q\,\mathrm{d}W_q, \qquad Y_{q_0} = Z_{q_0} = 0, \end{aligned} \tag{1}$$

where Y is the approximate forecast built from the explicit action $\bar{\Gamma}_q$, $Z = \psi - Y$ is its error, and $\psi = Y + Z$ is the exact forecast, a martingale. The inputs from Lecture 5 are:

A) *Global coercivity* (Lemma 12.1): $H_q(y)$ is comparable to $(K_q + \gamma y^2)^{-1}$ and $\|H_q(y)\|_{p\to p} \le (1+\varepsilon_q)/q$, $\varepsilon_q = C(g_q + g_q^2)$.

B) *The source* (Proposition 12.2): $b = b^{\mathrm{b}} + b^{1}$ with $\|b_q^{\mathrm{b}}\|_\infty \le C\,\gamma^{7/2}\,q^{-5/2}$ and $\|b_q^{1}(y)\|_2^2 \le C\gamma^4 q^{-5}\|t_q(y)\|_6^6$, where $t_q(y) = y\,1_{|y|>r_q}$.

C) *The coordinate equation* (12.5): $K_q(Y_q + \varrho_q(Y_q)) + \frac{\gamma}{3}Y_q^3 = K_q\Phi_q = P_q$, with a small Lipschitz correction, $\|D\,\varrho_q\|_{p\to p} \le \varepsilon_q$.

The goal is to show that Z is small and that Y rarely visits large fields, *uniformly in* N. This is UV stability. Neither equation can be estimated alone: the forward equation contains Z, and the source of the backward equation is controlled by the large fields of Y.

How a priori estimates for FBSDEs are usually obtained. For an FBSDE with Lipschitz coefficients one estimates $\mathbb{E}\|Z_q\|^2$ by Itô's formula and Gronwall, and closes either on a short time interval or under a monotonicity condition [MY99]. Here the time interval is infinite, the coefficients are not uniformly Lipschitz in the relevant norms, and the coupling is not small in a Lipschitz sense. What we have instead is structure:

- the backward equation is *linear* in Z, given Y, so it can be solved by a conditional representation formula with an explicit propagator;
- the forward equation is, after subtracting the Gaussian part, a cubic *elliptic* equation at each resolution, which is coercive and can be estimated pathwise;
- the two estimates are linked through two weighted energies whose weights are matched.

13 The energies

13.1 Moments at fixed cutoff

All estimates below are first proved on a finite interval $[q_0, T]$ and at fixed N. There, everything is finite.

Lemma 13.1. (Preliminary moments) *At fixed N and T, the maxima over $[q_0,T]$ of P, Y, ψ, Z have all moments, and so does $\int_{q_0}^{T}(\|\mathrm{Cov}_q\|_{\mathrm{HS}}^2 + \|\Xi_q\|_{\mathrm{HS}}^2)\mathrm{d}q$. Consequently the energies E_T, J_T and G_T defined below have all moments at fixed N, T.*

Proof. P has all maximal moments, since ν_N has all moments. Global convexity, $D^2\bar{\Psi}_q \ge c\,q$ and $D\bar{\Psi}_q(0) = 0$, gives $\|Y_q\| \le C\|P_q\|/q$. Doob's inequality for the martingale $\psi_q = \mathbb{E}_q\Phi_\infty$ gives its maximal moments, and $Z = \psi - Y$. For the integrands, $\|\mathrm{Cov}_q\|_{\mathrm{HS}} \le \mathrm{Tr}\mathrm{Cov}_q \le \mathbb{E}_q\|\Phi_\infty\|^2$, and $\|H_q(Y_q)\|_{\mathrm{HS}} \le C\sqrt{\dim H_N}/q$. The energies involve, besides Y and Z, the Gaussian field X and $\varrho_q(Y_q)$; finite-dimensional norm equivalence, $\|\varrho_q(y)\| \le \varepsilon_q\|y\|$ and Gaussian moments of X give their moments. □

These constants depend on N and T. They are used only to justify the absorptions below, never in the final bounds.

13.2 Subtracting the Gaussian part

The Gaussian part of the forecast is the free field at resolution q driven by the *same* innovation,

$$X_q := \int_{q_0}^{q} C_s \, \mathrm{d}W_s, \qquad \mathrm{Law}(X_q) = \mu^{Q_q}. \tag{13.1}$$

This is the formula (5.3) of Lecture 3, but W is the innovation of the interacting problem; X is a Gaussian process under the interacting measure. Since $\mathrm{d}(K_q X_q) = X_q \, \mathrm{d}q + \mathrm{d}W_q$,

$$K_q X_q = W_q + \int_{q_0}^{q} X_s \, \mathrm{d}s. \tag{13.2}$$

On the other hand $P_q = W_q + \int_{q_0}^{q} \psi_s \mathrm{d}s = W_q + \int_{q_0}^{q} (Y_s + Z_s) \mathrm{d}s$. Insert this into the coordinate equation (C) and subtract (13.2). With

$$V_q := Y_q + \varrho_q(Y_q) - X_q, \qquad F_q^Z := \int_{q_0}^{q} Z_s \, \mathrm{d}s, \tag{13.3}$$

the noise W cancels (V_q is not to be confused with the potentials V_t, V_N of Lectures 4–5) and we obtain the identity

$$K_q V_q + \frac{\gamma}{3} Y_q^3 = \int_{q_0}^{q} V_s \, \mathrm{d}s - \int_{q_0}^{q} \varrho_s(Y_s) \, \mathrm{d}s + F_q^Z, \qquad Y_q = V_q + X_q - \varrho_q(Y_q). \tag{13.4}$$

This is the equation we estimate. At each q it is the cubic elliptic equation $(q - \Delta) V + \frac{\gamma}{3}(V + X - \varrho)^3 = f$, a nonlinear Helmholtz equation in which the Gaussian field X enters, up to the small correction ϱ, only as a shift inside the cube, and the right side has a memory. Three remarks:

- the memory term has *coefficient one*: $\int_{q_0}^{q} V_s \mathrm{d}s$ is of the same size as the mass term $q V_q$. It cannot be treated as a forcing; it involves V itself, and the estimate must absorb it;
- the only trace of the backward equation is $F^Z = \int Z$;
- the identity is pathwise: no expectation has been taken.

13.3 Thresholds and energies

Define the signed excess of V over three quarters of the threshold, the large-field part of Y, and the large Gaussian values,

$$w_q = \mathrm{sgn}(V_q) \left(|V_q| - \tfrac{3}{4} r_q \right)_+, \qquad t_q = Y_q \mathbf{1}_{|Y_q| > r_q}, \qquad \xi_q = X_q \mathbf{1}_{|X_q| > r_q/32}, \tag{13.5}$$

all pointwise in space, with $r_q = (q/\gamma)^{1/2}$. The three random energies are

$$\begin{aligned} E_T &= \frac{1}{\gamma} \int_{q_0}^{T} q^{1/2} \|Z_q\|_2^2 \, \mathrm{d}q && \text{(backward energy)}, \\ J_T &= \int_{q_0}^{T} q^{-3/2} [q \|w_q\|_4^4 + \gamma \|w_q\|_6^6 + \gamma \|t_q\|_6^6] \mathrm{d}q && \text{(large-field action)}, \\ G_T &= \int_{q_0}^{T} [q^{-1/2} \|\xi_q\|_4^4 + \gamma q^{-3/2} \|\xi_q\|_6^6] \mathrm{d}q && \text{(Gaussian tails)}. \end{aligned} \tag{13.6}$$

Only *thresholded* sixth powers are integrated over all scales: $\mathbb{E}\|X_q\|_6^6 \approx q^{3/2}$ for $2 q_0 \le q \lesssim N^2$, so the unthresholded Gaussian contribution to J would grow like γN^2, not uniformly in N. The weights are matched to each other and to the source bounds:

quantity	weight	reason
$\|Z_q\|_2^2$	$q^{1/2}$	Hardy: $\int q^{-3/2}\|F^Z\|^2 \le 16\int q^{1/2}\|Z\|^2$
$\gamma\|t_q\|_6^6$	$q^{-3/2}$	source: $q^{7/2}\|b^1\|^2 \le C\gamma^3\cdot\gamma q^{-3/2}\|t\|_6^6$
$\|b^1_q\|_2^2$	$q^{7/2}$	backward propagator s/q and $\int_q^\infty s^{-3/2}\mathrm{d}s = 2q^{-1/2}$

13.4 Two Hardy inequalities

Lemma 13.2. (Hardy) *For f on $[q_0,\infty)$ with values in a Banach space, extended by zero to $(0,q_0)$,*

$$\Big|q^{-1}\int_0^q f_s\,\mathrm{d}s\Big|_{L^4(q^{-1/2}\mathrm{d}q)} \le \frac{8}{7}\|f\|_{L^4(q^{-1/2}\mathrm{d}q)}, \qquad \int_0^\infty q^{-3/2}\Big|\int_0^q f_s\,\mathrm{d}s\Big|^2\mathrm{d}q \le 16\int_0^\infty q^{1/2}\|f_q\|^2\mathrm{d}q. \tag{13.7}$$

Proof. Write $q^{-1}\int_0^q f_s\mathrm{d}s = \int_0^1 f_{q\theta}\mathrm{d}\theta$ and apply Minkowski's inequality. For the first bound, the substitution $u = q\,\theta$ gives $\|f_{\cdot\theta}\|_{L^4(q^{-1/2}\mathrm{d}q)} = \theta^{-1/8}\|f\|_{L^4(q^{-1/2}\mathrm{d}q)}$, and $\int_0^1\theta^{-1/8}\mathrm{d}\theta = \frac{8}{7}$. For the second, the left side is $\|q^{-1}\int_0^q f\|^2_{L^2(q^{1/2}\mathrm{d}q)}$, $\|f_{\cdot\theta}\|_{L^2(q^{1/2}\mathrm{d}q)} = \theta^{-3/4}\|f\|_{L^2(q^{1/2}\mathrm{d}q)}$, and $(\int_0^1\theta^{-3/4}\mathrm{d}\theta)^2 = 16$. □

The first inequality will estimate the memory term of (13.4). Its constant $\frac{8}{7}$ is *larger* than one. Coercivity from the mass term alone, which has coefficient one, would therefore not absorb the memory. The missing margin will come from the cubic term: at large fields it adds $\frac{1}{4}q|w|^4$ to the mass, and $1+\frac{1}{4}-\frac{8}{7} = \frac{3}{28} > 0$.

Exercises

Exercise 13.1. (Matched weights) Check the three weights in the table. Show that the forward estimate needs a weight at least $q^{1/2}$ on $\|Z\|^2$: test the second Hardy inequality with weight $q^{1/2-\sigma}$, $\sigma>0$, on $f = 1_{[a,2a]}$, $a\to\infty$. Show that the backward estimate of Section 15 also holds with weight $q^{1/2+\sigma}$ for $0<\sigma<\frac{1}{2}$. Which term breaks at $\sigma=\frac{1}{2}$, and how does the feedback coefficient depend on σ?

Exercise 13.2. (Gaussian sixth powers) Show that $\mathbb{E}\|X_q\|_6^6 = 15\big(\int_{q_0}^q a_2\big)^3 \le C\,q^{3/2}$, and that it is $\ge c\,q^{3/2}$ for $2\,q_0 \le q \le N^2$. Show that $\mathbb{E}\|\xi_q\|_6^6 \le C\,q^{3/2}\exp(-c\sqrt{q}/\gamma)$.

14 Forward coercivity

14.1 The scalar model

On one site the Laplacian disappears. We also drop the small Lipschitz correction ϱ (restored in Proposition 14.3) and allow an arbitrary continuous X, thinking of one site of the lattice, where $\mathrm{Var}X_q \approx \sqrt{q}$. Then (13.4) becomes a scalar equation with memory. All the mechanisms of the forward estimate are already present, and the constants can be checked by hand.

Lemma 14.1. (Scalar forward coercivity) *Let $\gamma>0$, $q_0<T$, let $X, f:[q_0,T]\to\mathbb{R}$ be continuous, and let v be continuous and solve*

$$q\,v_q + \frac{\gamma}{3}(v_q+X_q)^3 = \int_{q_0}^q v_s\,\mathrm{d}s + f_q, \qquad q\in[q_0,T]. \tag{14.1}$$

Define r_q, w_q, ξ_q as in (13.5), with $V=v$. Then

$$\frac{3}{28}\int_{q_0}^T q^{-1/2}w_q^4\,\mathrm{d}q + \frac{\gamma}{16}\int_{q_0}^T q^{-3/2}w_q^6\,\mathrm{d}q \le \frac{4225}{6}\gamma\int_{q_0}^T q^{-3/2}\xi_q^6\,\mathrm{d}q + \frac{12}{\gamma}\int_{q_0}^T q^{-3/2}f_q^2\,\mathrm{d}q. \tag{14.2}$$

Moreover the large-field part $t_q = (v_q + X_q)\,1_{|v_q+X_q|>r_q}$ *satisfies* $|t_q| \le \frac{32}{7}(|w_q| + |\xi_q|)$.

Proof. *Step 1: test with* w^3. Multiply (14.1) by w_q^3. We estimate the four terms where $w_q \neq 0$; there $|v_q| = |w_q| + \frac{3}{4} r_q$ and v_q, w_q have the same sign.

Mass. $q\, v_q w_q^3 = q|w_q|^4 + \frac{3}{4} q\, r_q |w_q|^3$.

Cube. $|X_q| \le \frac{r_q}{32} + |\xi_q|$, so $\operatorname{sgn}(w_q)\,(v_q + X_q) \ge |w_q| + \frac{23}{32} r_q - |\xi_q| \ge |w_q| + \frac{11}{16} r_q - |\xi_q|$ (we give up $\frac{1}{32} r_q$ to leave room, on the lattice, for the shift $\|\varrho_q^0\|_\infty \le \varepsilon_q r_q$, $\varepsilon_q \le \frac{1}{32}$). Since the cube is increasing (also if the right side is negative),

$$\frac{\gamma}{3} w_q^3 (v_q + X_q)^3 \ge \frac{\gamma}{3} |w_q|^3 \Big(|w_q| + \frac{11}{16} r_q - |\xi_q| \Big)^3 \ge \frac{\gamma}{12} w_q^6 + \frac{q}{4} w_q^4 - \frac{4225}{6} \gamma\, \xi_q^6,$$

by the elementary inequality (14.3) below and $\gamma r_q^2 = q$. *This is the large-field coercivity: at large fields the cube adds a sixth power and a quarter of the mass.*

Memory. Since $|v_s - w_s| \le \frac{3}{4} r_s$ and $\int_{q_0}^q r_s \mathrm{d}s \le \frac{2}{3} q\, r_q$,

$$w_q^3 \int_{q_0}^q v_s \,\mathrm{d}s \le \frac{3}{4} \cdot \frac{2}{3}\, q\, r_q |w_q|^3 + w_q^3 \int_{q_0}^q w_s \,\mathrm{d}s.$$

The first term, $\frac{1}{2} q\, r_q |w_q|^3$, is absorbed by the second mass term $\frac{3}{4} q\, r_q |w_q|^3$; the leftover $\frac{1}{4} q\, r_q |w_q|^3 \ge 0$ is dropped here, and is used on the lattice to absorb the memory of ϱ^0.

Forcing. By Young's inequality $w_q^3 f_q \le \frac{\gamma}{48} w_q^6 + \frac{12}{\gamma} f_q^2$.

Step 2: integrate. Collecting, at every q,

$$\frac{5}{4} q\, w_q^4 + \Big(\frac{\gamma}{12} - \frac{\gamma}{48} \Big) w_q^6 \le w_q^3 \int_{q_0}^q w_s \,\mathrm{d}s + \frac{4225}{6} \gamma\, \xi_q^6 + \frac{12}{\gamma} f_q^2.$$

Multiply by $q^{-3/2}$ and integrate over $[q_0, T]$. With $A = \int_{q_0}^T q^{-1/2} w_q^4 \mathrm{d}q$, Hölder and the first Hardy inequality (with w extended by zero outside $[q_0, T]$) give

$$\int_{q_0}^T q^{-1/2} |w_q|^3 \Big| q^{-1} \int_{q_0}^q w_s \,\mathrm{d}s \Big| \mathrm{d}q \le A^{3/4} \cdot \frac{8}{7} A^{1/4} = \frac{8}{7} A.$$

Since $\frac{5}{4} - \frac{8}{7} = \frac{3}{28}$, (14.2) follows. For the last claim, $|v + X| \le |w| + \frac{3}{4} r + \frac{r}{32} + |\xi|$, so on $\{|v + X| > r\}$ we have $\frac{7}{32} |v + X| \le |v + X| - \frac{25}{32} r \le |w| + |\xi|$. □

Lemma 14.2. (Two elementary inequalities) *For* $x, \rho \ge 0$, $(x - \rho)^3 \ge \frac{3}{4} x^3 - 65 \rho^3$. *For* $u, \rho, r \ge 0$,

$$\frac{1}{3} u^3 \Big(u + \frac{11}{16} r - \rho \Big)^3 \ge \frac{1}{12} u^6 + \frac{1}{4} r^2 u^4 - \frac{4225}{6} \rho^6. \tag{14.3}$$

Proof. $\frac{1}{4} x^3 + 64 \rho^3 - 3 x^2 \rho = \frac{1}{4} (x - 8\rho)^2 (x + 4\rho) \ge 0$, so $(x - \rho)^3 \ge x^3 - 3x^2\rho - \rho^3 \ge \frac{3}{4} x^3 - 65 \rho^3$. Apply it with $x = u + \frac{11}{16} r$: $\frac{1}{3} u^3 (x - \rho)^3 \ge \frac{1}{4} u^3 x^3 - \frac{65}{3} u^3 \rho^3$, and $u^3 x^3 \ge u^6 + 3(\frac{11}{16})^2 r^2 u^4$ with $\frac{3}{4}(\frac{11}{16})^2 = \frac{363}{1024} > \frac{1}{4}$. Finally $\frac{65}{3} u^3 \rho^3 \le \frac{1}{6} u^6 + \frac{4225}{6} \rho^6$, since $\frac{1}{6} (u^3 - 65\rho^3)^2 \ge 0$, and $\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$. □

What the scalar lemma says. The excess of the forecast over the threshold is controlled by two things only: how often the *Gaussian* field is large, which is a known quantity (the Gaussian-tail functional G_T, with $\|G_T\|_{L^p} \le C \gamma^3 p^5$ independently of N), exponentially small at large q, and the *backward* forcing f. The memory term, although of the same size as the mass term, is absorbed thanks to the extra quartic damping supplied by the cube at large fields. There is no smallness in the coupling: the estimate is uniform in γ once the threshold scales like $r_q = (q/\gamma)^{1/2}$. And it is pathwise.

14.2 The lattice version

Proposition 14.3. (Forward coercivity [FBSDE26]) *For q_0 large (depending on γ), at every cutoff N and every T, pathwise,*

$$J_T \le C\,(G_T + E_T), \qquad \|G_T\|_{L^p(\mathbb{P})} \le C\gamma^3 p^5 \quad (p \ge 1), \tag{14.4}$$

with C independent of N, T and p.

Proof: the changes from the scalar case. Test (13.4) with w_q^3 in $L^2(\Lambda_N)$ and proceed as in Lemma 14.1, with Hölder's inequality in space as well as in scale. Three new terms appear.

The Laplacian. $\langle w^3, -\Delta_N V\rangle \ge 0$. Indeed $w^3 = \phi'(V)$ for the convex function $\phi(v) = \frac{1}{4}(|v| - \frac{3}{4}r)_+^4$, and for any nondecreasing ϕ',

$$\langle \phi'(V), -\Delta_N V\rangle = N^{-1}\sum_{x\sim x'} (\phi'(V(x)) - \phi'(V(x')))\,(V(x) - V(x')) \ge 0,$$

summing over unordered nearest-neighbour pairs. So the elliptic term only helps.

The coordinate correction. Split $\varrho_q(y) = \varrho_q^0 + \varrho_q^1$ according to $|y| \le r_q$ or not (writing $\varrho_q(y) = \mathsf{L}_q(y)\,y$ with L the averaged derivative). Then $\|\varrho_q^0\|_\infty \le \varepsilon_q r_q$ shifts the threshold by a small amount, and $\|\varrho_q^1\|_j \le \varepsilon_q \|t_q\|_j$, $j = 4, 6$. If $\varepsilon_q \le \frac{1}{32}$, then $|Y| \le |w| + \frac{13}{16}r_q + |\xi| + |\varrho^1|$; on $\{|Y| > r_q\}$ this gives $\frac{3}{16}|Y| \le |w| + |\xi| + |\varrho^1|$, so $\|t_q\|_j \le \frac{16}{3}(\|w_q\|_j + \|\xi_q\|_j + \varepsilon_q\|t_q\|_j)$, and for small ε

$$\|t_q\|_j \le C\,(\|w_q\|_j + \|\xi_q\|_j), \qquad \|\varrho_q^1\|_j \le C\,\varepsilon_q\,(\|w_q\|_j + \|\xi_q\|_j).$$

This also supplies the t part of J_T. These are *operator* bounds: ϱ is nonlocal, but small in every L^p. In the cube the error is $\rho = |\xi| + |\varrho^1|$, and the cube error $\frac{4225}{6}\gamma\|\rho\|_6^6$ is at most $C\gamma(\|\xi_q\|_6^6 + \varepsilon_q^6\|w_q\|_6^6)$: the w part is absorbed for small ε, the ξ part goes into G_T. The memory of ϱ^0 costs at most $\frac{2}{3}(\sup_s \varepsilon_s)\; q\, r_q\|w_q\|_3^3$ and is absorbed by the spare $\frac{1}{4}q\,r_q\|w_q\|_3^3$ when $\sup_s \varepsilon_s < \frac{3}{8}$ (automatic, since $\varepsilon \le \frac{1}{32}$); the memory of ϱ^1 costs $C\sup_q \varepsilon_q\,(A + G_T)$, with $A = \int_{q_0}^T q^{-1/2}\|w_q\|_4^4 \mathrm{d}q$, by Hölder, Hardy and Young, and is absorbed by the spare $\frac{3}{28}$ if $C\sup_q \varepsilon_q < \frac{3}{56}$; the subtraction is legitimate because $A < \infty$ pathwise at fixed N, T, by continuity on $[q_0, T]$. All are deterministic smallness conditions on g_{q_0}, independent of p.

The backward forcing. $|\langle w^3, F^Z\rangle| \le \frac{\gamma}{48}\|w\|_6^6 + \frac{12}{\gamma}\|F^Z\|_2^2$, and by the second Hardy inequality $\gamma^{-1}\int q^{-3/2}\|F_q^Z\|_2^2 \mathrm{d}q \le 16\; E_T$.

Gaussian tails. $X_q(x)$ is centred Gaussian with variance $\int_{q_0}^q a_2 \le C\sqrt{q}$, while the threshold is $r_q/32 = \frac{1}{32}(q/\gamma)^{1/2}$. By Minkowski's inequality in x and translation invariance of $\mathrm{Law}(X_q)$, $|||\xi_q||_j^j|_{L^p(\mathbb{P})} \le \|\xi_q(0)|^j|_{L^p(\mathbb{P})}$, and by Hölder, for $j = 4, 6$,

$$|||\xi_q||_j^j|_{L^p(\mathbb{P})} \le (\mathbb{E}|X_q(0)|^{2jp})^{\frac{1}{2p}}\,\mathbb{P}\,(|X_q(0)| > r_q/32)^{\frac{1}{2p}} \le C_j p^{j/2} q^{j/4} \mathrm{e}^{-c\sqrt{q}/(p\gamma)},$$

and integrating in q with $u = \sqrt{q}/(p\gamma)$ gives $\|G_T\|_p \le C\gamma^3 p^5$. □

No information on the interacting field has been used, apart from the equation itself: the Gaussian tails are tails of the free field X, which is Gaussian under the interacting measure because it is built from the innovation.

Exercises

Exercise 14.1. (Without the cube) Replace $\frac{\gamma}{3}(v + X)^3$ by 0 in (14.1), and show that the argument fails: find v solving $q\,v_q = \int_{q_0}^q v_s \mathrm{d}s + f_q$ on $[q_0, \infty)$ with $\int q^{-1/2} v^4 = \infty$ for f with $\int q^{-3/2} f^2 < \infty$ (try v constant). Testing with v^3 gives $(1 - \frac{8}{7})\int q^{-1/2} v^4 \le \ldots$, which is empty. What replaces the missing margin, and the missing w^6 term that absorbs the forcing, when the cube is present?

Exercise 14.2. (**Discrete convexity**) Prove that $\langle \phi'(u), -\Delta_N u\rangle \geq 0$ for every convex $\phi \in C^1$ and $u \in H_N$, and that equality holds iff $\phi'(u(x)) = \phi'(u(x'))$ on every edge $x \sim x'$ with $u(x) \neq u(x')$.

Exercise 14.3. (**Simulation**) Solve (14.1) numerically with X a scalar Brownian motion run with clock $\int_{q_0}^{q} s^{-1/2}\mathrm{d}s$ (the three-dimensional bubble $a_2(s) \approx s^{-1/2}$), e.g. $\gamma = 1$, $q_0 = 1$, $T = 10^4$. With $f = 0$ the solution typically stays below the threshold (it can cross briefly at small q, where $|X_q|$ is comparable to r_q), so the left side is zero or small; the right side does not, because at these parameters $g_{q_0} = 1$ is not small and $|X_q| > r_q/32$ on most scales. Add a forcing pulse $f_q = \kappa\, q\, r_q\, 1_{[a,2a]}(q)$ with κ of order one, so that v crosses $\frac{3}{4} r_q$. Compare the left side of (14.2) with its f-term and its ξ-term separately, as functions of κ and a; then decrease γ (e.g. $\gamma = 10^{-2}$) and observe the ξ-term becoming negligible at large q.

15 Backward feedback and closure

15.1 The backward representation

The backward equation in (1) is linear in Z. Its propagator is the adapted operator solution of

$$\partial_s \mathcal{P}_{q,s} = \mathcal{P}_{q,s} H_s(Y_s), \qquad \mathcal{P}_{q,q} = I, \qquad s \geq q. \tag{15.1}$$

By (A), $\|\mathcal{P}_{q,s}\|_{2\to 2} \leq \exp\int_q^s \frac{1+\varepsilon_u}{u}\mathrm{d}u \leq L_0\,\frac{s}{q}$, with $L_0 = \exp\left(\int_{q_0}^{\infty} \varepsilon_u \frac{\mathrm{d}u}{u}\right) \leq \exp\left(C\,(g_{q_0} + g_{q_0}^2)\right)$.

Lemma 15.1. (**Conditional representation**) *For $q_0 \leq q \leq T$, at fixed N,*

$$Z_q = \mathbb{E}_q\Big[\,\mathcal{P}_{q,T} Z_T + \int_q^T \mathcal{P}_{q,s}\, b_s(Y_s)\,\mathrm{d}s\Big]. \tag{15.2}$$

Proof. By the product rule and (1), $\mathrm{d}_s(\mathcal{P}_{q,s} Z_s) = \mathcal{P}_{q,s} H_s Z_s \mathrm{d}s - \mathcal{P}_{q,s}(b_s + H_s Z_s)\mathrm{d}s + \mathcal{P}_{q,s}\Xi_s \mathrm{d}W_s = -\mathcal{P}_{q,s} b_s \mathrm{d}s + \mathcal{P}_{q,s}\Xi_s \mathrm{d}W_s$. The stochastic integral is a true martingale by Lemma 13.1 and the deterministic bound on $\mathcal{P}$. Integrate from q to T and take $\mathbb{E}_q$. □

So Z is the expected source accumulated on the unresolved scales, transported back to q, plus the boundary error. This is the conditional-expectation form of a backward equation, as in $F_t = \mathbb{E}_t[F_T]$ for the force (Theorem 7.4); here the source is the defect.

15.2 Three contributions

Split $Z = Z^{\mathrm{t}} + Z^{\mathrm{b}} + Z^{\mathrm{l}}$ according to the terminal term and the two parts of the source. We estimate each in the scalar manner: norms of the propagator times norms of the source.

Bounded fields. By (B),

$$\|Z_q^{\mathrm{b}}\|_2 \leq \int_q^{\infty} L_0 \frac{s}{q}\, C\gamma^{7/2} s^{-5/2}\,\mathrm{d}s = 2L_0 C\gamma^{7/2} q^{-3/2}, \qquad \frac{1}{\gamma}\int_{q_0}^{\infty} q^{1/2}\|Z_q^{\mathrm{b}}\|_2^2\,\mathrm{d}q \leq C\gamma^6 q_0^{-3/2}.$$

This is where the two correction steps of Lecture 5 are used: with one step the source would be $O(s^{-2})$ and the first integral would diverge logarithmically.

Boundary. $q\|Z_q^{\mathrm{t}}\|_2 \leq L_0\ \mathbb{E}_q\|T Z_T\|_2$, so, with $\delta_{N,T,p} := \|T Z_T\|^2_{L^{2p}(\mathbb{P};L^2)}$,

$$\left|\frac{1}{\gamma}\int_{q_0}^{T} q^{1/2}\|Z_q^{\mathrm{t}}\|_2^2 \mathrm{d}q\right|_{L^p(\mathbb{P})} \leq \frac{L_0^2}{\gamma}\int_{q_0}^{\infty} q^{-3/2}\mathrm{d}q\ \ \delta_{N,T,p} = \frac{2L_0^2}{\gamma q_0^{1/2}}\ \delta_{N,T,p}.$$

At fixed N, $\delta_{N,T,p} \to 0$ as $T \to \infty$ (Lemma 16.1).

Large fields. Let $A^1 := \int_{q_0}^{T} s^{7/2}\|b_s^1(Y_s)\|_2^2 \mathrm{d}s \leq C\gamma^4 \int s^{-3/2}\|t_s\|_6^6 \leq C\gamma^3 J_T$ by (B). By Cauchy–Schwarz in s,

$$\|Z_q^1\|_2 \leq \frac{L_0}{q}\ \mathbb{E}_q \int_q^T s\|b_s^1\|_2 \mathrm{d}s \leq \frac{L_0}{q}\Big(\int_q^{\infty} s^{-3/2}\mathrm{d}s\Big)^{1/2}\,\mathbb{E}_q[(A^1)^{1/2}],$$

so that, by Jensen's inequality, $\|Z^1_q\|_2^2 \le 2L_0^2 q^{-5/2}\,\mathbb{E}_q A^1$. Integrating, and using that $\mathbb{E}_q$ is a contraction on $L^p(\mathbb{P})$,

$$\left\| \frac{1}{\gamma}\int_{q_0}^{T} q^{1/2}\|Z^1_q\|_2^2 \mathrm{d}q \right\|_{L^p} \le \frac{2L_0^2}{\gamma}\int_{q_0}^{\infty} q^{-2}\mathrm{d}q\,\|A^1\|_{L^p} \le \frac{C\gamma^2}{q_0}\|J_T\|_{L^p}.$$

This is how the clipping error is controlled. The large-field source b^1 is not small; but its squared L^2 norm is controlled by the sixth power of the large-field part, $\|b^1_q\|_2^2 \le C\gamma^4 q^{-5}\|t_q\|_6^6$ (B), and after transport by the propagator and integration with the energy weight it contributes to the backward energy only through J_T, with the small factor γ^2/q_0.

Proposition 15.2. (Backward feedback) *At every cutoff,*

$$\|E_T\|_{L^p} \le C\gamma^6 q_0^{-3/2} + C\gamma^{-1} q_0^{-1/2}\,\delta_{N,T,p} + C\gamma^2 q_0^{-1}\|J_T\|_{L^p}, \tag{15.3}$$

with C independent of N, T and p.

Proof. $\|Z\|^2 \le 3\,(\|Z^{\mathrm{t}}\|^2 + \|Z^{\mathrm{b}}\|^2 + \|Z^{\mathrm{l}}\|^2)$ and the three estimates above. □

15.3 Closing the loop

Theorem 15.3. (Cutoff-uniform energy estimate) *There is q_0, depending only on γ, such that for every N, T and $p \ge 1$,*

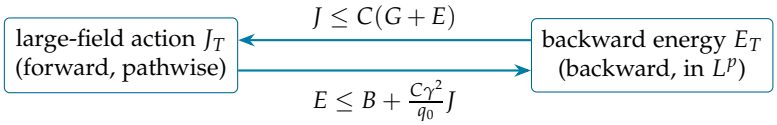
$$\|E_T\|_{L^p} + \|J_T\|_{L^p} \le C_\gamma\,(1 + p^5 + \delta_{N,T,p}). \tag{15.4}$$

Proof. Let $x = \|E_T\|_p$ and $y = \|J_T\|_p$. By Propositions 14.3 and 15.2,

$$y \le A_0\,(C\gamma^3 p^5 + x), \qquad x \le B_0 + B_1\,\delta_{N,T,p} + \eta\,y, \qquad \eta = \frac{C\gamma^2}{q_0},$$

with A_0, B_0, B_1 and the constant in η independent of N, T, p, and bounded for all q_0 beyond the thresholds already required. Choose q_0 such that also $A_0\eta \le \frac{1}{2}$. Then $y \le A_0(C\gamma^3 p^5 + B_0 + B_1\,\delta_{N,T,p}) + \frac{1}{2}y$. Since $y < \infty$ by Lemma 13.1, it can be absorbed, and x follows. □

The structure of the argument is the point:



Forward coercivity estimates the large fields using the backward energy; the backward representation returns that energy with a small coefficient. Neither half closes on its own. The smallness is in the *feedback*, γ^2/q_0, not in the coupling of either equation, and it is independent of the moment order, so a single q_0 closes the loop for every p. The growth p^5 is then inherited from the Gaussian tails G_T, and gives stretched exponential moments $\mathbb{E}\exp\{c\,(E_\infty + J_\infty)^{1/5}\} < \infty$ once the boundary term is removed. Note that $\gamma^2/q_0 = g^2_{q_0}$.

15.4 Consequences

The boundary term $\delta_{N,T,p}$ is removed in Lecture 7, at each fixed N, by letting $T \to \infty$. Assuming this for the moment, we obtain

$$\sup_N \|E^N_\infty\|_{L^p} + \sup_N \|J^N_\infty\|_{L^p} \le C_\gamma\,(1 + p^5), \qquad \sup_N \mathbb{E}\|Z^N_q\|_2^2 \le C_\gamma q^{-5/2}. \tag{15.5}$$

The second bound follows from the representation formula on $[q,\infty)$: the three contributions are, up to the factor 3, $O(q^{-3})$, 0 and, by the computation above, $2L_0^2 q^{-5/2}\,\mathbb{E}\,A^1$. So the approximate forecast differs from the exact one by $O(q^{-5/4})$ in $L^2(\mathbb{P};L^2)$ (root mean square), uniformly in the lattice. Since $\|H_q\| \le C/q$, the coupling term $H_q Z_q$ in the forward equation is $O(q^{-9/4})$ in $L^2(\mathbb{P};L^2)$, integrable in q: the forward equation is, up to an integrable drift, the explicit SDE $\mathrm{d}Y = b(Y)\mathrm{d}q + H(Y)\mathrm{d}W$.

Remark 15.4. (**Comparison with the variational method**) In the Boué–Dupuis approach (Theorem 9.1) one estimates a drift u which is the costate up to $\dot{Q}^{1/2}$, and the renormalisation is a change of variables $u = -4\lambda\dot{Q}^{1/2}{:}X^3{:}+r$ (in the normalisation $\lambda\int{:}\varphi^4{:}$ of Lecture 4) [BG20]. Here the explicit action $\bar{\Gamma}$ plays the role of all such changes of variables at once, in the forecast coordinate, and Z plays the role of the remainder r. The coercivity of the quartic term is used in both, but here it is used *pathwise in the forward equation*, while the remainder is estimated in expectation through the backward equation.

Exercises

Exercise 15.1. (**The scalar backward estimate**) On one site, with $H_q(y) = (q+\gamma y^2)^{-1}$, show that

$$\mathcal{P}_{q,s} = \frac{s}{q}\exp\left(-\int_q^s \frac{\gamma Y_u^2}{u\,(u+\gamma Y_u^2)}\mathrm{d}u\right) \le \frac{s}{q}.$$

In which sense do large fields make the backward equation more contractive? Redo the three estimates of Section 15 on one site.

Exercise 15.2. (**Absorption needs finiteness**) Give an example of nonnegative numbers x,y (possibly infinite) with $y \le 1+x$, $x \le 1+\frac{1}{2}y$ and $y=\infty$. Where is Lemma 13.1 used in the proof of Theorem 15.3?

Exercise 15.3. (**Moment growth**) From $\|E_\infty + J_\infty\|_p \le C(1+p^5)$ for all $p \ge 1$, deduce $\mathbb{E}\exp\{c\,(E_\infty+J_\infty)^{1/5}\} < \infty$ for small c.

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